The Wiig KM Cycle

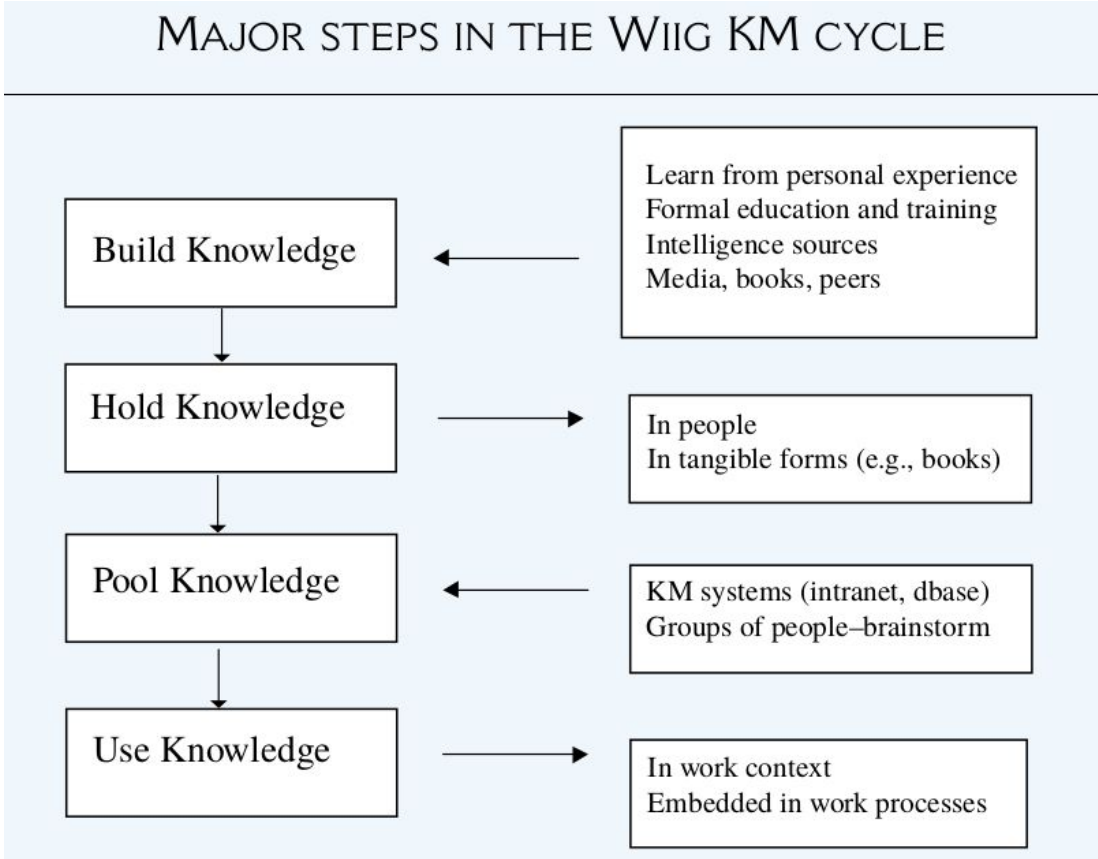
Introduction

- Wiig (1993) focuses on the **three conditions** that need to be **present for a organization** to conduct its business successfully:
  - **it must have a business (products/services) and customers;**
  - **it must have resources (people, capital, and facilities); and**
  - **it must have the ability to act.**
- The **third point is emphasized in the Wiig KM cycle**.
- **Knowledge is the principal force** that determines and drives the ability to act intelligently.
- Wiig identifies the **major purpose of KM** as **an effort "to make the enterprise intelligent-acting by facilitating the creation, cumulation, deployment and use of quality knowledge"**

The Wiig KM Cycle

Wiig's KM cycle **addresses how knowledge is built and used** as individuals or as organizations. There are **four major steps** in this cycle, as shown in Figure

- **Building knowledge.**
- **Holding knowledge.**
- **Pooling knowledge.**
- **Applying knowledge.**



The Wiig KM Cycle (cont…)

- **Although the steps are shown to be independent and sequential**, But **we may perform some of the functions and activities in parallel**.
- It is **also possible to cycle back to repeat functions and activities** performed earlier **but with a different emphasis and/or level of detail.**
- The cycle **focuses on identifying and relating the functions and activities** that we engage in to make products and services as knowledge workers.

Step-1: Build Knowledge

Building knowledge refers to activities ranging from market research to focus groups, surveys, competitive intelligence, and data mining applications.

Building knowledge consists of **five major activities**:

- **Obtain knowledge (Knowledge Creation).**
- **Analyze knowledge (Knowledge Analysis).**
- **Reconstruct/synthesize knowledge (Knowledge Synthesis).**
- **Codify and model knowledge (Knowledge codification).**
- **Organize knowledge (Knowledge Organization).**

Step-1: Build Knowledge (Knowledge Creation)

- **Knowledge Creation** may occur through **R&D projects**, **innovations by individuals** to improve the way they perform their tasks, **experimentation**, **reasoning with existing knowledge**, and **hiring of new people**.
- **Knowledge may also be created through knowledge importing** (e.g., **elicit knowledge from experts and from procedure manuals, engage in joint ventures to obtain technology, or transfer people between departments**).
- Finally, **knowledge may be created through observing the real world** (e.g., **making site visits, observing processes after the introduction of a change**).

Step-1: Build Knowledge (Knowledge Analysis)

Knowledge Analysis consists of:

- **Extracting** what appears to be knowledge from obtained material (e.g., analyze transcripts and identify themes, listen to an explanation, and select concepts for further consideration).
- **Abstracting** extracted materials (e.g., form a model or a theory).
- **Identifying** patterns extracted (e.g., trend analysis).
- **Explaining** relations between knowledge fragments (e.g., compare and contrast, causal relations).
- **Verifying** that extracted materials correspond to the meaning of original sources (e.g., meaning has not been corrupted through summarizing, collating, and so on).

Step-1: Build Knowledge (Knowledge Synthesis)

Knowledge Synthesis or **Reconstruction** consists of **generalizing analyzed material to obtain broader principles, generating hypotheses to explain observations, establishing conformance between new and existing knowledge** (e.g. **corroborating validity in light of what is already known), and updating the total knowledge pool by incorporating the new knowledge.**

Step-1: Build Knowledge (Knowledge Codification)

Codifying and modeling knowledge involves **how we represent knowledge in our minds** (mental models, for example), **how we then assemble the knowledge into a coherent model**, **how we document the knowledge in books and manuals**, and **how we encode it** in order **to post it to a knowledge repository**.

Step-1: Build Knowledge (Knowledge Organization)

Finally, **knowledge is organized for specific uses** and **according to an established organizational framework** (such as standards and categories).

Examples include a help desk service or a list of frequently asked questions (FAQs) on the company intranet.

This **organization is usually done using some form of knowledge ontology (conceptual model) and taxonomy (classification rules)**.

Examples include an official list of key words or categories, knowledge object attribute specifications, and guidelines for translation.

Step-2: Hold Knowledge

Holding Knowledge consists of remembering, accumulating knowledge in repositories, embedding knowledge in repositories, and archiving knowledge.

- **Remembering Knowledge** means that the individual has retained or remembered that item of knowledge (i.e., knowledge has been internalized or understood by a given individual).
- **Accumulating Knowledge** in a repository means that a computer-resident knowledge base has been created and that knowledge has been encoded, permitting it to be stored in organizational memory.
- **Embedding Knowledge** consists of ensuring that it is part of business procedures (e.g., added to a procedures manual or training course).
- **Archiving Knowledge** involves creating a scientific library and systematically retiring out-of-date, false, or no longer relevant knowledge from the active repository.
- **Archiving typically involves storing the content in another, less costly, or less bulky medium for less frequent future retrieval.**

Step-2: Hold Knowledge (cont…)

Examples of knowledge held by companies include **intellectual property, patents, knowledge documented in the form of research reports, technical papers, or tacit knowledge, which remains in the minds of individuals but which may be elicited and embedded in the knowledge base or repository** (e.g., tips, tricks of the trade, case studies, videotapes of demonstrations by experts, and task support systems).

In this way, the organization's holdings of valuable knowledge are documented in repositories or in people and are therefore available for future reference and use.

Step-3: Pool Knowledge

Knowledge Pooling consists of coordinating, assembling, and accessing and retrieving knowledge.

- **Coordination of Knowledge** typically requires the formation of collaborative teams to work with particular content in order to create a “who knows what” network.
- Once knowledge sources are identified, they are then **Assembled** into background references for a library or repository in order to facilitate subsequent access and retrieval.
- **Access and Retrieval** then addresses being able to consult with knowledgeable people about difficult problems, obtaining a second opinion from an expert, or discussing a difficult case with a peer.
- **Knowledge can be accessed and retrieved directly from the repository as well (e.g., using a knowledge base system to obtain advice on how to do something or reading a knowledge document in order to arrive at a decision).**

Step-3: Pool Knowledge (cont…)

Organizations may pool knowledge in a variety of ways.

The employee who does not have the necessary knowledge and know-how to solve a particular problem can contact others in the organization who have solved similar problems either by obtaining the information from the organizational knowledge repository or by finding an expert through the expertise locator network and contacting that person directly.

The individual can then organize all this information and request that more experienced knowledge workers validate the content.

Step-4: Use Knowledge

There are innumerable ways to apply the knowledge, including the following:

- **Use established knowledge to perform a routine task**—for example, make standard products, provide a standard service, or use the expert network to find out who is knowledgeable about a particular area.
- **Use general knowledge to survey exceptional situations at hand**— for example, determine what the problem is and estimate potential consequences.
- **Use knowledge to describe the situation and scope of the problem**—for example, identify the problem and show generally how to handle it.
- **Select relevant special knowledge to handle the situation**—for example, identify who you need to consult with or want to address the problem.

Step-4: Use Knowledge (cont…)

- **Observe and characterize the situation with special knowledge**—for example, make a comparison with known patterns, take a history, and collect and organize required information to act.
- **Analyze the situation with knowledge**—for example, judge whether it can be handled internally or whether outside help will be required.
- **Synthesize alternative solutions with knowledge**—for example, identify options and outline possible approaches.
- **Evaluate potential alternatives using special knowledge**—for example, determine the risks and benefits of each possible approach.
- **Use knowledge to decide what to do**—for example, rank alternatives, select one, and do a reality check.
- **Implement the selected alternative**—for example, execute the task and authorize the team to proceed.

Conclusion

- When knowledge is applied to work objects, routine and standard tasks are approached in a different way than difficult or unusual tasks.
- Routine or standard tasks are typically carried out using “compiled” knowledge that we can readily access and use almost unconsciously or automatically.
- Difficult tasks are usually performed in a more deliberate and conscious manner, for knowledge workers cannot use automated knowledge in unanticipated situations.

Summary of the key WIIG KM cycle activities

Obtain
Analyze
Reconstruct
Synthesize
Codify
Model
Organize

Remember
Accumulate in repositories
Embed in repositories
Archive

Coordinate
Assemble
Reconstruct
Synthesize
Access
Retrieve

Perform tasks
Survey, describe
Select
Observe, analyze
Synthesize
Evaluate
Decide
Implement

References

- Wiig, K. (1993). Knowledge management foundations. Arlington, TX: Schema Press.
- Dalkir, K. (2005). Knowledge Management In Theory And Practice. McGill University: Elsevier Butterworth–Heinemann.